

HAVANA CASABLANCA, Cuba

WMO: 783250

Lat: 23.144N

Lon: 82.342W

Elev: 50

StdP: 100.73

Time Zone: -5.00 (CUB)

Period: 90-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.9	(c) 17.2	(d) 7.6	(e) 6.5	(f) 18.4	(g) 9.6	(h) 7.5	(i) 19.1	(j) 11.1	(k) 23.3	(l) 10.3	(m) 22.7	(n) 3.3	(o) 30	(p) 0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.9	(c) 32.2	(d) 25.3	(e) 31.6	(f) 25.3	(g) 31.1	(h) 25.3	(i) 26.7	(j) 30.3	(k) 26.4	(l) 30.0	(m) 26.1	(n) 29.7	(o) 4.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.1	28.9	25.3	20.6	28.6	25.1	20.3	28.4	83.7	30.4	82.3	30.1	81.1	29.9	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.1	(c) 8.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 10.7	(b) 9.1	(c) 8.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.4	22.7	22.9	23.1	25.0	26.2	27.4	27.8	28.1	27.6	26.5	24.5	22.9		
	DBStd	2.57	2.01	2.16	2.22	1.61	1.15	1.04	0.80	0.86	0.97	1.33	1.65	2.04		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	4	1	1	1	0	0	0	0	0	0	0	0	1		
	CDD10.0	5619	393	361	406	451	503	521	552	561	527	512	435	399		
	CDD18.3	2582	136	128	148	201	244	271	293	302	277	253	185	143		
	CDH23.3	21204	514	599	777	1384	2010	2707	3159	3346	2817	2219	1091	581		
	CDH26.7	5702	53	93	133	289	464	813	1067	1186	870	513	163	58		
(14) Wind	WSAvg	3.4	4.0	4.3	4.3	4.3	3.7	3.4	2.6	2.4	2.3	3.0	3.3	3.8		
(15) Precipitation	PrecAvg	1177	75	65	41	66	97	170	107	108	147	162	91	66		
	PrecMax	1884	310	226	235	210	331	694	283	265	422	524	426	226		
	PrecMin	663	0	2	0	0	0	12	17	15	27	5	6	0		
	PrecStd	288	68	48	40	57	79	126	69	56	84	103	77	55		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.5	30.9	31.3	32.2	31.9	33.0	32.4	33.2	33.3	32.1	31.1	29.8		
		MCWB	22.6	22.3	22.8	23.8	25.3	26.0	25.3	25.6	25.6	25.2	24.6	23.2		
	2%	DB	28.1	29.2	29.7	30.4	30.4	31.6	31.5	31.8	31.6	30.7	29.2	28.1		
		MCWB	22.3	22.5	22.8	23.6	24.4	25.6	25.4	25.6	25.4	25.4	23.9	22.8		
	5%	DB	26.9	27.5	28.0	29.2	29.7	30.7	30.9	31.3	30.9	29.8	28.3	26.9		
		MCWB	22.3	22.2	22.5	23.1	24.1	25.5	25.6	25.6	25.4	25.2	23.8	22.5		
	10%	DB	25.9	26.4	26.9	28.3	29.0	30.1	30.5	30.9	30.3	29.1	27.5	26.0		
		MCWB	21.9	21.9	21.9	22.8	23.7	25.4	25.5	25.6	25.3	24.8	23.3	22.0		
	0.4%	WB	24.1	24.2	24.5	25.2	26.3	27.1	27.1	27.2	26.7	26.9	25.3	24.4		
		MCDB	27.2	27.4	28.5	29.4	29.9	30.5	30.5	30.6	30.3	29.9	28.4	27.2		
	2%	WB	23.5	23.5	23.8	24.5	25.6	26.6	26.5	26.7	26.2	26.2	25.0	23.7		
		MCDB	26.4	26.6	27.1	28.5	29.0	30.0	30.1	30.4	30.0	29.3	28.1	26.4		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	23.0	23.0	23.4	24.0	25.0	26.2	26.1	26.3	26.0	25.7	24.5	23.2		
		MCDB	25.7	26.0	26.7	27.7	28.3	29.6	29.8	30.0	29.7	28.7	27.3	26.0		
	10%	WB	22.5	22.6	22.8	23.5	24.5	25.8	25.8	26.0	25.7	25.3	24.0	22.7		
		MCDB	25.1	25.5	26.1	27.1	27.8	29.1	29.5	29.7	29.3	28.2	26.5	25.4		
	5% DB	MDBR	6.0	6.5	6.4	6.2	5.6	5.6	5.7	5.9	5.8	5.1	5.2	5.6		
		MCDBR	7.7	8.5	8.3	7.6	6.5	6.4	6.2	6.4	6.7	6.2	6.7	7.1		
(36) Mean Daily Temperature Range	5% WB	MCWBR	3.1	3.4	3.4	2.8	2.8	2.7	2.7	2.5	2.5	2.4	2.6	3.1		
		MCDBR	6.7	7.4	7.0	6.9	5.9	6.0	5.8	6.0	6.3	5.8	6.1	6.3		
	5% WB	MCWBR	3.2	3.4	3.4	2.9	2.9	3.0	2.9	2.7	2.8	2.6	2.9	3.2		
		MCDBR	3.2	3.4	3.4	2.9	2.9	3.0	2.9	2.7	2.8	2.6	2.9	3.2		
(40) Clear-Sky Solar Irradiance	taub		0.362	0.370	0.378	0.401	0.426	0.447	0.454	0.431	0.416	0.407	0.377	0.372		
	taud		2.528	2.511	2.472	2.410	2.380	2.377	2.347	2.434	2.467	2.471	2.505	2.521		
	Ebn at Noon		904	918	924	902	871	846	840	863	875	869	880	878		
	Edh at Noon		95	103	111	120	123	123	127	116	110	105	97	93		
(44) All-Sky Solar Radiation	RadAvg		3.90	4.77	5.81	6.51	6.42	5.98	6.47	6.17	5.44	4.68	4.04	3.70		
	RadStd		0.31	0.33	0.21	0.23	0.61	0.44	0.30	0.21	0.28	0.33	0.19	0.30		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	+0.89	N/A	N/A	N/A	N/A	N/A	N/A	-11	N/A	N/A

Nomenclature: See separate page